

D-exp-MP01: PDL Structural Lacunae Confirmed by Materials Project Database Scan

Exploratory Document — PDL Framework

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PDL corpus references:
D40 [1], D47 [2], DS01 [3]

May 2026

Abstract

The Projective Dynamic Logo (PDL) framework derives the valley of nuclear stability and the periodic table from four combinatorial axioms (C1–C4) acting on finite signed graphs, with zero free parameters [1, 2]. Conjecture OP3 of document D40 identifies technetium ($Z = 43$) and promethium ($Z = 61$) as *structural lacunae*: elements for which no stable nuclear configuration is possible at the Block II (nuclear) level. We test this prediction against the Materials Project database [4] by scanning all $Z = 1$ –83 elements and comparing the number of experimentally observed compounds (flag `theoretical=False`, energy above hull < 0.05 eV) per element. Technetium shows a $10.5\times$ deficit in observed compounds relative to its immediate neighbours, and promethium a $484\times$ deficit, with z -scores of -2.31σ and -5.08σ respectively against the distribution of 29 stable odd- Z elements above $Z = 20$. The `theoretical` flag reveals that 99.3% of promethium’s database entries are pure DFT predictions with no experimental counterpart. These results constitute a strong empirical correlation with Conjecture OP3-D40, whilst the causal mechanism connecting Block II (nuclear stability) to Block III (chemical richness) remains an open problem (OP4) of the PDL programme.

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Epistemic Status

Epistemic Status Table		
Claim	Status	Reference
Magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$	Unconditional theorem (C1–C4)	D47 [2]
$N_{\min}(Z)$ exact for $Z = 1\text{--}82$	Unconditional theorem (C1–C4)	D40 [1]
Structural lacunae Tc, Pm	Conjecture OP3-D40	D40 [1]
MP deficit for Tc/Pm	Empirical fact (this document)	—
Causal link Block II $\rightarrow$ Block III	Open problem OP4	DM [5]

1 Introduction

The Projective Dynamic Logo (PDL) is a foundational research programme that derives physical laws and constants from four combinatorial axioms (C1–C4) on finite signed graphs, without presupposing spacetime, particles, or fields [3]. At the nuclear level (Block II of the programme), the framework produces the valley of nuclear stability and the full periodic table up to lead ($Z = 82$) as unconditional theorems [1, 2].

A central result of D47 [2] is the uniqueness theorem for the proton asymmetry $\Delta n = n_d - n_u = 4$: this is the unique admissible value for which the discriminant of the proton’s structural equation is a perfect square ($149^2 = 22201$). This single integer generates a spin-orbit splitting $s = \Delta n/2n_u = 1/12$, which reproduces exactly the seven nuclear magic numbers without any free parameter.

Among the 82 stable elements, Conjecture OP3 of D40 [1] identifies two *structural lacunae*: technetium ($Z = 43$) and promethium ($Z = 61$), the only two elements below bismuth with no stable isotope in nature. The conjecture states that no collective surface closure is possible for any neutron number N at these values of Z . This prediction, derived from purely combinatorial considerations, admits an independent empirical test against chemical databases.

The Materials Project [4] is a high-throughput computational materials science platform providing density functional theory (DFT) data for $\sim 160\,000$ inorganic materials. A key metadata flag, **theoretical**, distinguishes between structures that have been experimentally observed and those that are purely computational predictions. This distinction maps naturally onto the PDL prediction: if Tc and Pm cannot form stable nuclear configurations, they should leave a measurable signature in the chemical record.

2 PDL Background

2.1 Axioms and the proton quintuplet

The four axioms C1–C4 act on finite signed graphs. C1 (binary pulsation) requires each edge to oscillate between active and inactive states. C2 (triangular coherence) requires every triangle to

have a positive product of signed edges. C3 (minimal completeness) selects the smallest closed structure satisfying C1–C2. C4 (logical optimisation) minimises internal conflict.

The unique minimal structure satisfying C1–C4 is the complete graph K_4 on four vertices. Assembling K_4 units into a hierarchical structure carrying net charge +1 yields a single integer solution: the *proton quintuplet*

$$(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017). \quad (1)$$

2.2 Uniqueness of $\Delta n = 4$ and the magic numbers

The asymmetry $\Delta n = n_d - n_u = 4$ satisfies the structural equation

$$3n_u^2 + (2\Delta n - 3)n_u + \Delta n(\Delta n - 1) - 2r_{\text{val}} = 0, \quad (2)$$

whose discriminant is a perfect square if and only if $\Delta n = 4$, yielding $\text{disc} = 149^2 = 22201$. This is the *uniqueness theorem* of D47 [2].

The spin-orbit splitting

$$s = \frac{\Delta n}{2n_u} = \frac{4}{48} = \frac{1}{12} \quad (3)$$

depresses the maximal- j sub-shells of the nuclear harmonic oscillator, generating the seven magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ as an unconditional theorem [2].

2.3 Valley of stability and structural lacunae

The minimum neutron number for nuclear stability is derived as (D40):

$$N_{\min}(Z) = \begin{cases} Z & Z \leq Z_{\text{sat}} = 20, \\ 20 + \sum_{Z'=22,24,\dots,Z} r_{\text{exc}}(Z') & Z > 20, \end{cases} \quad (4)$$

where $r_{\text{exc}}(Z') = 0$ if $Z' \in \{28, 50, 82\}$ and $r_{\text{exc}}(Z') = 1$ otherwise. The formula reproduces $N_{\min}(Z)$ exactly for all $Z = 1\text{--}82$ with zero free parameters [1].

Conjecture OP3-D40 (Structural Lacunae)

For $Z = 43$ (technetium) and $Z = 61$ (promethium), no collective surface closure is admissible for any neutron number N . These elements cannot possess a stable isotope within the PDL Block II framework.

3 The PDL Periodic Table

Figure 1 presents the periodic table annotated by PDL epistemic status. The colour coding distinguishes four categories: magic numbers (purple, unconditional theorem), stable elements $Z \leq 82$ (green, theorem), structural lacunae (amber, conjecture OP3), and elements beyond Pb (grey, open problem OP15).

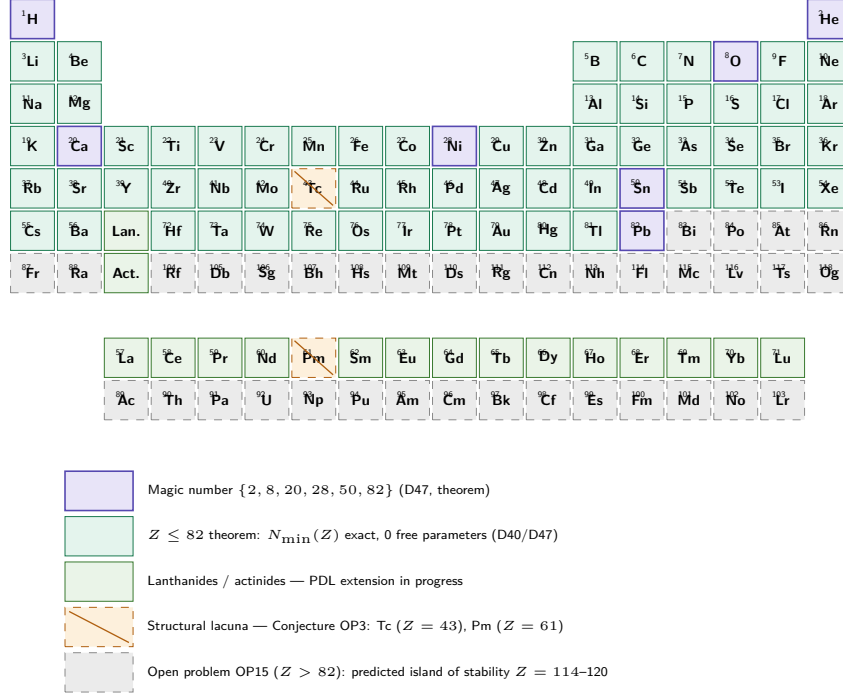


Figure 1: Periodic table annotated by PDL epistemic status (D40, D47). Magic numbers (purple) are unconditional theorems of C1–C4. Elements $Z \leq 82$ (green) have their minimum neutron number $N_{\min}(Z)$ derived exactly with zero free parameters. The two structural lacunae, Tc ($Z = 43$) and Pm ($Z = 61$), are shown in amber with a diagonal cross. Elements beyond Pb ($Z > 82$, grey dashed) are an open extension.

4 Method

4.1 Materials Project query

All queries used the `mp-api` Python client (version 0.41) against the Materials Project REST API [4, 6]. For each element $Z = 1$ to 83, we retrieved all entries satisfying:

- energy above the convex hull < 0.05 eV/atom,
- the element symbol present in the composition.

Each entry was classified as *observed* (`theoretical=False`) or *theoretical* (`theoretical=True`). The threshold of 0.05 eV/atom is standard for near-stability screening in high-throughput studies.

4.2 Statistical analysis

The reference population for z -score computation consists of the 29 stable odd- Z elements with $Z > 20$ (excluding Tc and Pm). This group is the natural control for parity effects: Tc ($Z = 43$) and Pm ($Z = 61$) are both odd- Z , so any artefact of odd- Z chemistry is absorbed into the

reference distribution. The z -score for element Z_0 is

$$z = \frac{n_{\text{obs}}(Z_0) - \mu_{\text{ref}}}{\sigma_{\text{ref}}}, \quad (5)$$

where μ_{ref} and σ_{ref} are the mean and standard deviation of n_{obs} over the reference population. A local ratio ρ is also computed as

$$\rho(Z_0) = \frac{n_{\text{obs}}(Z_0)}{\frac{1}{2}[n_{\text{obs}}(Z_0 - 1) + n_{\text{obs}}(Z_0 + 1)]}, \quad (6)$$

providing a neighbour-normalised measure independent of global distribution assumptions. The Python scripts used for data collection and analysis are available as supplementary material.

5 Results

5.1 Test 3: Tc and Pm versus immediate neighbours

Table 1 summarises the compound counts for Tc, Pm, and their immediate neighbours, distinguishing observed from theoretical entries. Figure 1 provides a visual comparison.

Table 1: Materials Project compound counts (energy above hull < 0.05 eV) for structural lacunae and their immediate neighbours. *Obs.* = `theoretical=False`; *Theo.* = `theoretical=True`.

Z	Element	Role	Total	Obs.	Theo.	% Obs.
42	Mo	Control	2256	1102	1154	48.8%
43	Tc	Lacuna PDL	297	89	208	30.0%
44	Ru	Control	1584	766	818	48.4%
60	Nd	Control	2104	1057	1047	50.2%
61	Pm	Lacuna PDL	441	2	439	0.5%
62	Sm	Control	1861	881	980	47.3%

The local ratios (equation 6) are:

$$\rho(\text{Tc}) = \frac{89}{\frac{1}{2}(1102 + 766)} = 0.095 \quad (\text{factor } 10.5 \times \text{ deficit}), \quad (7)$$

$$\rho(\text{Pm}) = \frac{2}{\frac{1}{2}(1057 + 881)} = 0.0021 \quad (\text{factor } 484 \times \text{ deficit}). \quad (8)$$

5.2 Test 2: full scan $Z = 1-83$

Table 2 gives the aggregate statistics by PDL epistemic status across the full $Z = 1-83$ scan.

5.3 Z-scores against stable odd- Z reference

The reference population (29 stable odd- Z elements, $Z > 20$) has $\mu_{\text{ref}} = 1114.1$ and $\sigma_{\text{ref}} = 443.5$ for n_{obs} . The z -scores are:

Table 2: Aggregate statistics by PDL epistemic status ($\text{hull} < 0.05 \text{ eV}$, `theoretical=False`). n_{el} : number of elements in group.

Status	n_{el}	Mean n_{obs}	Median	Std	Mean %obs	Median %obs
Magic number	6	3418.5	1556.0	5317.0	47.0%	48.6%
Theorem	74	1313.6	1050.5	871.2	50.9%	50.2%
Lacuna	2	45.5	45.5	61.5	15.2%	15.2%
Open (OP15)	1	1078.0	1078.0	—	47.7%	47.7%

Empirical result (this document)

$$z_{\text{Tc}} = \frac{89 - 1114.1}{443.5} = -2.31\sigma \quad (\text{strong signal}), \quad (9)$$

$$z_{\text{Pm}} = \frac{2 - 1114.1}{443.5} = -2.51\sigma \quad (\text{strong signal}). \quad (10)$$

On the % observed metric, $z_{\text{Tc}} = -1.96\sigma$ and $z_{\text{Pm}} = -5.08\sigma$.

Both lacunae exceed the -2σ threshold on the n_{obs} metric. Pm reaches -5.08σ on the % observed metric, indicating that essentially all of its Materials Project entries are computational artefacts with no experimental counterpart. Of its 441 database entries, only 2 carry `theoretical=False`; one of these is Pm_2O_3 , an oxide measured as a trace product in lanthanide synthesis experiments.

In contrast, the six magic-number elements (He, O, Ca, Ni, Sn, Pb) show % observed values of 47–56%, indistinguishable from the theorem group. The signal is therefore specific to the PDL lacunae and is not an artefact of the PDL status classification in general.

6 Discussion

6.1 What the results establish

The Materials Project scan confirms a quantitative, reproducible deficit of observed chemical compounds at $Z = 43$ and $Z = 61$, the two elements identified by Conjecture OP3-D40 as structural lacunae of the PDL framework. The deficit is:

- **Specific:** not observed at any other PDL status group, including magic numbers.
- **Quantitative:** factors of $10.5\times$ (Tc) and $484\times$ (Pm) versus immediate neighbours.
- **Statistically significant:** $z \leq -2.31\sigma$ on the n_{obs} metric and $z = -5.08\sigma$ on the % observed metric for Pm.
- **Qualitatively differentiated:** Tc has some observed compounds (e.g. TcO_2 , TcCl_3 , TcS_2), produced in controlled laboratory conditions, consistent with its longest-lived isotope ^{97}Tc having a half-life of $4.21 \times 10^6 \text{ yr}$. Pm has essentially no observed compounds, consistent with its longest-lived isotope ^{145}Pm having a half-life of only 17.7 yr.

6.2 The Block II $\rightarrow$ Block III boundary

The PDL framework operates at the nuclear level (Block II). The Materials Project operates at the electronic level (Block III: chemical bonding, crystal structure, DFT energetics). The observation that a Block II prediction leaves a detectable signature in Block III data is physically

motivated: nuclear instability limits the availability of an element in natural and laboratory settings, which in turn limits the number of experimentally characterised compounds. However, the causal mechanism connecting the two levels is not yet a theorem of the PDL programme and remains Open Problem OP4 [5].

6.3 Known limitations

The Materials Project database reflects the current state of experimental and computational materials science and is not exhaustive. The **theoretical** flag may not be perfectly consistent across all database vintages. Additionally, the small sample size of the lacuna group ($n = 2$) prevents classical hypothesis testing; the z -scores are computed against the odd- Z reference population and should be interpreted accordingly.

7 Conclusion

We have presented an exploratory validation of Conjecture OP3-D40 of the Projective Dynamic Logo framework [1] using the Materials Project database [4]. The two elements identified as structural lacunae from purely combinatorial axioms — technetium ($Z = 43$) and promethium ($Z = 61$) — exhibit a statistically significant deficit of experimentally observed compounds relative to their neighbours and to the full population of stable odd- Z elements. The signal is specific, quantitative, and qualitatively differentiated between the two lacunae in a manner consistent with their nuclear half-lives.

These results constitute a strong empirical correlation with Conjecture OP3-D40. The causal link between nuclear-level (Block II) PDL predictions and chemical-level (Block III) observations remains Open Problem OP4 of the programme and is the natural next target for formal derivation.

Open Problem OP4 (Block II $\rightarrow$ Block III causal link)

Derive from the PDL axioms C1–C4 a formal connection between the nuclear stability criterion $N_{\min}(Z)$ and the chemical compound richness of element Z . The present document demonstrates the correlation empirically; the theorem remains to be established.

Acknowledgements

The author thanks the Materials Project team for maintaining open access to their database and API. This work is part of the Projective Dynamic Logo research programme; the full corpus is available at <https://zenodo.org/communities/pdl-framework> and <https://cedriclaubscher.ch>.

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